

Machine Learning-Based Classification of Space Debris and Satellites Using Orbital Parameters

Sushnato Majhi

Electrical Engineering

National Institute of Technology, Jamshedpur

Jamshedpur, India

majhisushnato.tech@gmail.com

Kethavath Raghavendra Naik

Electrical Engineering

National Institute of Technology, Jamshedpur

Jamshedpur, India

raghavendra.ee@nitjsr.ac.in

Kumari Namrata

Electrical Engineering

National Institute of Technology, Jamshedpur National Institute of Technology, Jamshedpur

Jamshedpur, India

namrata.ee@nitjsr.ac.in

Veerpratap Meena

Electrical Engineering

National Institute of Technology, Jamshedpur

Jamshedpur, India

vmeena1@ee.iitr.ac.in

Devendra Gowda

Electrical Engineering

Acharya Institutes

Bangalore, India

devendragowdapatil@gmail.com

Abstract—The classification of space debris and active satellites is crucial for maintaining space situational awareness and preventing collisions. This study presents a machine learning approach that utilizes two-line element (TLE) data to distinguish between satellites and debris based on key orbital parameters, including eccentricity, inclination, mean motion, right ascension of the ascending node (RAAN), and semi-major axis. Two models—a feedforward neural network (FNN) and a random forest classifier—were developed and evaluated for their classification performance. The TLE-derived parameters were preprocessed to address data inconsistencies, with the feature importance analysis revealing the inclination and the semi-major axis as dominant discriminators. The Random Forest model utilized an ensemble of decision trees with Gini impurity criteria, while the FNN employed a three-layer architecture with ReLU activation and dropout regularization. The random forest classifier achieved superior accuracy (92.4%) compared to the FNN (87.1%), demonstrating robustness to noisy TLE data and non-linear feature relationships. Both models significantly outperformed traditional clustering-based taxonomic methods, particularly in handling edge cases such as debris with satellite-like orbits. This data-driven approach provides a scalable alternative to resource-intensive optical or radar-based classification systems. Future work will integrate the temporal evolution of orbital parameters to address decaying orbits and the detection of maneuvers.

Index Terms—Space debris classification, Two-Line Element data, orbital parameters, machine learning, random forest, and feedforward neural network.

I. INTRODUCTION

The increasing volume of space debris, ranging from defunct satellites to rocket fragments, threatens satellite operations and mission safety. Traditional classification methods, based on orbital dynamics and manual tracking, struggle with scalability. This study leverages Two-Line Element-derived orbital parameters and evaluates three machine learning models—FNN, gradient boosting, and random forest—for classifying space objects as operational satellites or debris. The results show the potential of AI to improve space situational

awareness, improve collision avoidance, and support long-term orbital sustainability.

In this research paper, a machine learning classification framework was created to differentiate between active satellites and space debris in Earth's orbit based on orbital parameters obtained from Two-Line Element (TLE) data. A dataset was obtained and pre-processed to eliminate outliers, and feature scaling was performed to maintain homogeneity in parameters. Key orbital characteristics—eccentricity, inclination, mean motion, semi-major axis, RAAN, and velocity—were chosen considering their physical significance and variability between debris and satellites. Three machine learning models were experimented with and compared: a feedforward neural network (FNN), gradient boosting classifier, and random forest, with random forest resulting in the highest accuracy of 99.6% and exhibiting high robustness to data imbalance. Feature importance analysis was also performed, and the orbital distributions were visualized to understand the behavior of the model. It was verified that debris accumulates in low Earth orbit with more chaotic dynamics than satellites. This work adds a scalable and data-driven method to increase space situational awareness and provides a foundation for the addition of other observation data and real-time observations.

The overview of the paper is as follows. The introduction presents the emerging issue of space debris and the confines of conventional classification techniques, which require the application of machine learning with TLE data. Section II presents related work, with an emphasis on AI-based detection technologies and global initiatives aimed at mitigating space debris. Section III presents the approach, specifically the TLE data preprocessing, feature design, and training of three machine learning models: FNN, Gradient Boosting, and Random Forest. Section IV explains the reasoning behind the choice of the most important orbital parameters and their physical significance in distinguishing between satellites and space debris. Section V presents a comparison of model

performance, indicating that the random forest delivers the best accuracy and stability. Section VI examines the distribution of orbital parameters through visualizations and identifies distinct behavioral differences between satellites and debris. Section VII concludes the work and outlines future directions, including combining light curve data, applying sophisticated machine learning (ML) techniques, and collaborating with space agencies to classify space objects in real-time.

II. RELATED WORK

Recent developments in the detection and collection of space debris highlight the need to resolve orbital congestion, which now represents a serious threat to working satellites and space vehicles. With more than 40,000 objects tracked around Earth, the risk of devastating collisions has never been greater. Detection techniques have changed dramatically with the incorporation of artificial intelligence (AI), including hybrid machine learning models with Bi-LSTM and CNN architectures [1]. These systems provide more than 99% accuracy in debris classification and radar cross-section size estimation with optimized computational efficiency for real-time applications. This precision is important to distinguish between working satellites and dangerous debris and to allow timely evasive actions.

Ground-based radar campaigns, such as Europe's twice-yearly beam-park exercises [2], now track hundreds of objects in one day, including debris from past breakup events like the Pegasus rocket explosion. Such campaigns offer important insight into the changing debris environment, and astronomers can use the results to forecast future collision threats and targets for removal. Meanwhile, future space-based sensors like the 2025 Space Debris Hunter satellite will look for debris as small as 1 cm in objects too small to be tracked by ground radars but too fast to avoid catastrophic collisions. This ability fills a key shortcoming in existing detection systems, which tend to fail to detect smaller debris that can nevertheless inflict substantial harm. Advancements such as tethered nets and AI-controlled spacecraft further improve capture consistency, with systems automatically making adjustments to debris spin or unorthodox motion [3]–[5]. These technologies are important in providing secure and effective removal operations since they are able to accommodate the unpredictable nature of debris in space. Policy infrastructures support these technological advances, including NASA's debris mitigation policy and ESA's Design for Removal (D4R) standards, which invite satellite manufacturers to design features like standardized docking ports for easier removal missions in the future. Incorporating such design practices into future satellites would help the space industry minimize the risk of future-generation debris and make it less challenging to remove spent spacecraft. In the future, sensor miniaturization and AI-assisted automation will change debris management. Today, AI-powered algorithms focus on high-risk objects and anticipate collision scenarios ahead of time, allowing pre-emptive mitigation. For example, AI can study the history to determine patterns of debris behavior and better predict possible collisions. With this predictive

ability, satellite operators can initiate evasive maneuvers long in advance, reducing the chances of damage considerably.

With advancing technologies, emphasis is on cost-effective, scalable solutions to maintain orbital environments for future missions. Integrating AI in detection and removal systems is likely to be a key to this purpose. AI has the capability to maximize resource utilization through optimal identification of the most dangerous debris and prioritizing their removal according to risk analysis. AI-driven automation can also simplify operations by minimizing the necessity of human intervention and reducing operational expenses.

Long-term viability of space operations relies on the capability to monitor and mitigate space debris efficiently. With the rising number of satellites in orbit due to the proliferation of satellite constellations and emerging new space missions, the likelihood of collision and resultant debris creation will rise unless preventive action is undertaken. Thus, continued development and research in detection and removal technologies are essential. These actions not only secure the present space infrastructure but also lay the ground for future space missions by ensuring a safe and accessible orbital environment. Through the application of AI, advanced materials, and novel capture mechanisms, the space industry is set to overcome the intricate problem of space debris, and space remains an open frontier for exploration and development [4]–[9].

III. METHODOLOGY

Two-line element data was sourced from publicly available satellite catalogs. The dataset included both active satellites and space debris, identified based on historical tracking information. The methodology of this study followed a structured approach beginning with data preprocessing, where the collected Two-Line Element data was cleaned to remove duplicate entries and anomalies. Feature scaling techniques were applied to normalize the dataset, ensuring uniformity across different parameters. Feature engineering was then conducted to analyze and visualize key orbital parameters, such as eccentricity, inclination, mean motion, semi-major axis, and right ascension of the ascending node (RAAN), allowing for an in-depth understanding of their variations between satellites and debris. Following this, three machine learning models, Feed-forward Neural Network (FNN) and Random Forest Classifier were trained using the processed dataset. Hyperparameter tuning was performed to optimize model performance, followed by an evaluation stage in which classification metrics, including accuracy, precision, recall, and F1-score, were analyzed. Finally, a comparative assessment of the models was conducted to determine the most effective classification technique and identify areas of improvement. Significant parameters influence the classification process [10], [11].

IV. FEATURE SELECTION

The selected features—eccentricity, inclination, mean motion, semi-major axis, RAAN, and velocity—were chosen

because they encapsulate the fundamental dynamics and geometry of an object's orbit, which tend to differ systematically between operational satellites and space debris.

Eccentricity was chosen because it reflects how circular or elliptical an orbit is. Operational satellites are typically placed in stable, near-circular orbits to ensure consistent performance and coverage. In contrast, debris often originates from explosions or collisions, leading to more erratic, elongated orbits.

Inclination describes the tilt of the orbit relative to Earth's equator. Satellites are launched into specific inclinations for mission-specific coverage (e.g., polar, equatorial, or sun-synchronous orbits). Debris, being uncontrolled, can have inclinations that vary more widely due to gravitational perturbations and past disruptive events.

Mean motion (the number of orbits per day) serves as an indirect measure of orbital altitude. Active satellites are placed at precise altitudes to maintain desired orbital periods, whereas debris may spread across a range of altitudes after fragmentation, affecting their mean motion.

The semi-major axis defines the size of the orbit and is closely related to both eccentricity and mean motion. Like mean motion, it helps distinguish mission-driven orbital planning from the more chaotic distribution of debris.

RAAN indicates the orientation of the orbit in space. Operational satellites often share similar RAAN values due to coordinated deployments in orbital planes (especially in constellations), while debris orbits tend to show more random orientations.

Velocity was included to capture dynamic differences, although its variation is relatively subtle. Still, in combination with the other features, it can contribute to distinguishing orbital energy states and support finer classification.

In summary, these parameters were selected not just for their availability in TLE data, but because they offer physical insight into how an object behaves in orbit—helping machine learning models detect patterns that separate well-managed, predictable satellite behavior from the more chaotic nature of space debris.

A. Eccentricity

Eccentricity measures the deviation of an orbit from a perfect circle and is given by:

$$e = \frac{r_a - r_p}{r_a + r_p} \quad (1)$$

Where r_a are the apoapsis distance and r_p the periapsis distance. Space debris often exhibits higher eccentricities due to past collisions or fragmentation events, while operational satellites tend to have lower, more stable eccentricities.

Inclination refers to the tilt of an object's orbit relative to the Earth's equatorial plane and is computed as:

$$i = \cos^{-1} \left(\frac{h_z}{|h|} \right) \quad (2)$$

where h_z is the z-component of the angular momentum vector and h is the magnitude of the angular momentum. Satellites

are generally placed in specific inclination ranges depending on their mission, whereas debris may have a wider range due to perturbations.

1) *Mean Motion*: Mean motion n represents the number of orbits an object completes in a day and is given by:

$$n = \sqrt{\frac{\mu}{a^3}} \quad (3)$$

where μ is the standard gravitational parameter of Earth and is the semi-major axis. Objects with higher mean motion values correspond to lower-altitude orbits, while higher-altitude objects have lower mean motion values.

2) *Semi-Major Axis*: The semi-major axis a defines the size of an orbit and is related to the orbital period by Kepler's Third Law:

$$T^2 = \frac{4\pi^2 a^3}{\mu} \quad (4)$$

where T is the orbital period. Satellites are strategically placed at specific altitudes, while debris can be found at various altitudes due to orbital perturbations.

3) *RAAN*: Ω describes the orientation of an orbit in space and is given by

$$\Omega = \tan^{-1} \left(\frac{h_x}{-h_y} \right) \quad (5)$$

where h_x and h_y are components of the angular momentum vector. Controlled satellites maintain relatively stable RAAN values, whereas debris may exhibit irregular variations due to external forces.

4) *Velocity*: Orbital velocity v depends on the position in the orbit and is computed using

$$v = \sqrt{\mu \left(\frac{2}{r} - \frac{1}{a} \right)} \quad (6)$$

where r is the radial distance from the Earth's center. Although velocity is crucial for orbital dynamics, our analysis found that variations in velocity between satellites and debris were relatively small.

B. Machine Learning Models

1) **Gradient Boosting Classifier (GBC)** : A gradient boosting model, as seen in Table I, was trained using the selected Two-Line Element features for classification. The model builds an ensemble of decision trees in a stage-wise manner, where each successive tree attempts to correct the errors of its predecessor. The predicted probability for class k is obtained through the softmax function applied to the cumulative outputs of the trees: [12].

$$\hat{y}_k = \frac{\exp(F_k(x))}{\sum_{j=1}^K \exp(F_j(x))} \quad (7)$$

where $F_k(x)$ is the score (logit) produced for class k by the ensemble of trees, and what K is the total number of classes.

The model is trained to minimize the *categorical cross-entropy loss*:

$$L = - \sum y_k \log(\hat{y}_k) \quad (8)$$

where y_k is the true label (one-hot encoded), and $\hat{y}_k$ is the predicted probability for class k . Each boosting iteration fits a new decision tree to the *negative gradient* of the loss function with respect to the model's current predictions, effectively correcting the residual errors as seen in Table I

TABLE I
CLASSIFICATION REPORT: GRADIENT BOOSTING

Class	Precision	Recall	F1-Score	Support
0	1.00	0.99	0.99	4186
1	0.92	0.99	0.95	529
Accuracy			0.99	4715
Macro Avg	0.96	0.99	0.97	4715
Weighted Avg	0.99	0.99	0.99	4715

2) *Feed-forward Neural Network (FNN)*: An FNN model was trained with the selected Two-Line Element features. The architecture included multiple hidden layers with ReLU activation and a softmax output layer for classification. The forward propagation is defined as: [13].

$$a^{(l)} = f \left(W^{(l)} a^{(l-1)} + b^{(l)} \right) \quad (9)$$

where W^l are the weights and biases in the layer, a^{l-1} and is the activation function. The loss function used was categorical cross-entropy.

$$L = - \sum y \log(\hat{y}) \quad (10)$$

where y is the true label and $\hat{y}$ is the predicted probability as we see in Table II

TABLE II
CLASSIFICATION REPORT: FEEDFORWARD NEURAL NETWORK (FNN)

Class	Precision	Recall	F1-Score	Support
0	0.89	1.00	0.94	4186
1	0.00	0.00	0.00	529
Accuracy			0.89	4715
Macro Avg	0.44	0.50	0.47	4715
Weighted Avg	0.79	0.89	0.84	4715

3) *Random Forest Classifier*: A random forest model was implemented using multiple decision trees, which is visualized in Table III. The final classification is determined by majority voting among individual trees. Each decision tree follows the rule:

$$y = \arg \max \sum_{i=1}^n P_i(y|X) \quad (11)$$

where $P_i(y | X)$ is the probability of class y given input X from the i^{th} trees. Feature importance was analyzed using the Gini index:

$$G = 1 - \sum p_i^2 \quad (12)$$

where p_i is the probability of class at a given node .

TABLE III
CLASSIFICATION REPORT: RANDOM FOREST

Class	Precision	Recall	F1-Score	Support
0	1.00	1.00	1.00	4186
1	0.98	0.98	0.98	529
Accuracy			1.00	4715
Macro Avg	0.99	0.99	0.99	4715
Weighted Avg	1.00	1.00	1.00	4715

V. MODEL EVALUATION

TABLE IV
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision (Label 0)	Recall (Label 0)
Neural Network	0.887805	0.887805	1.000000
Random Forest	0.995758	0.997611	0.997611
Gradient Boosting	0.988759	0.998552	0.988772

Table IV provides us with the comparison of the three models compared. Below are the three models compared neural network, random forest, and gradient boosting—the best-performing model was the random forest model overall, with an exceptionally high accuracy of 99.58% accompanied by almost perfect precision, recall, and F1-scores for both classes. Gradient Boosting also performed well, just behind Random Forest with an accuracy of 98.88%, and with similarly high scores, especially having a balance between precision and recall for both classes. In contrast, the neural network model showed a significantly lower accuracy of 88.78%, performing well for the majority class (Label 0) with perfect recall but completely failing to identify the minority class (Label 1)—resulting in zero precision, recall, and F1-score for that class. This shows that although random forest and gradient boosting are very accurate for both classes, the neural network has difficulty with class imbalance and gives poor generalization for minority class prediction.

VI. RESULTS AND DISCUSSION

A. Distribution of Semi-Major Axis

In Fig. 1, the semi-major axis is shown. The semi-major axis distribution indicates a high density of both active satellites and debris at lower values (approximately 10,000 km). This suggests that the majority of objects, such as active satellites and debris, are in low Earth orbit (LEO), which is an area highly used for communication, Earth observation, and other satellite uses. As the semi-major axis goes above 10,000 km, the number declines considerably, implying fewer objects in more elevated orbits such as medium Earth orbit (MEO) or geostationary orbit (GEO). Debris is more numerous than active satellites at lower semi-major axis values, pointing towards the gathering of space debris in LEO.

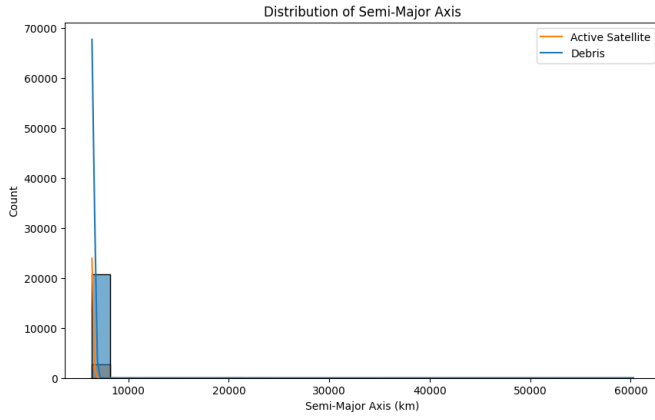


Fig. 1. Semi-Major Axis

B. Distribution of RAAN (RAAN)

In Fig. 2, the RAAN is illustrated. Active satellites (blue) are the predominant distribution of RAAN for all values, with peaks at regular intervals. Debris (red) occurs less frequently but is ubiquitous. Active satellites are more uniformly distributed over orbital planes as result of being placed on purpose for functional reasons. Debris, by contrast, shows random patterns of distribution since it is not controlled. Pinnacles in RAAN for operational satellites imply clustering within certain orbital planes, presumably determined by mission needs or satellite constellations.

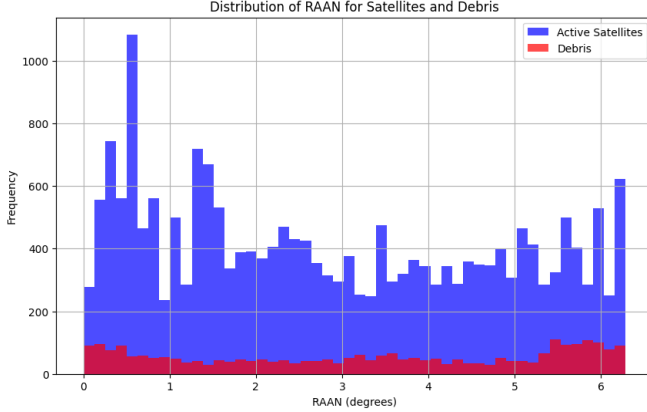


Fig. 2. Characteristics of Right Ascension of the Ascending Node of Satellites and debris

C. Distribution of Mean Motion

Fig. 3 indicates the mean motion. The distribution of mean motion indicates that debris prevails at higher points (approximately 0.07 revolutions per minute) and active satellites have a lower spread around similar points. Greater mean motion is associated with shorter orbital periods that are typical for objects in lower orbits. The prevalence of debris at these values once again indicates its concentration in LEO. Active satellites possess a more guided and systematic mean motion distribution than debris.

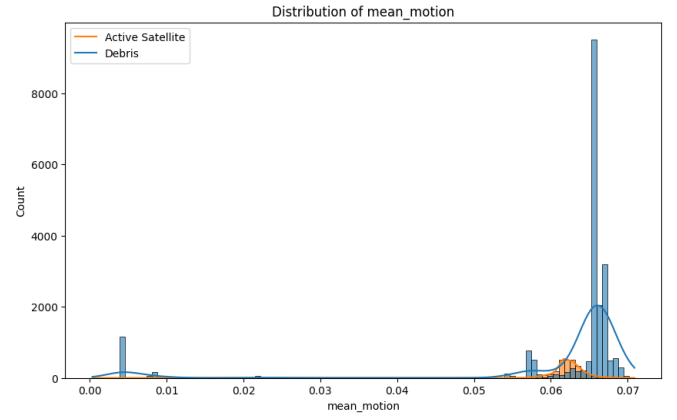


Fig. 3. Mean Motion Comparison of Satellites and Debris

D. Inclination vs Eccentricity

In Fig. 4, the connection between inclination and eccentricity is shown. Active satellites (orange) are grouped around low values of eccentricity and inclination, signifying stable and almost circular orbits. Debris (blue) is more variable, with objects having high eccentricity and inclination. Active satellites are strategically placed in stable orbits to reduce perturbations and guarantee mission success. Debris exhibits chaotic orbital parameters due to fragmentation events or collisions that alter their trajectories unpredictably. Debris dispersal at greater eccentricities and inclinations points to the difficulties presented by unregulated space bodies.

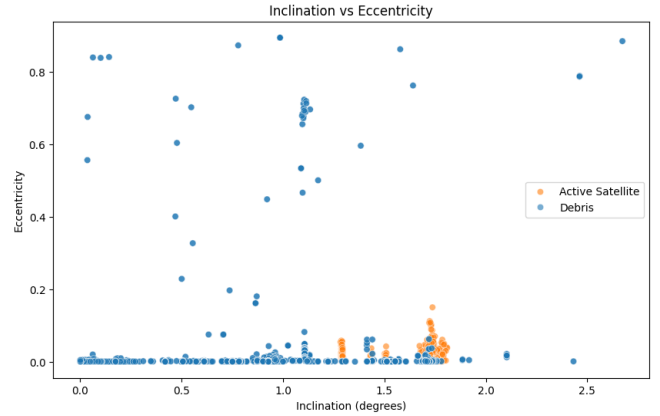


Fig. 4. Eccentricity vs Inclination Comparison of Satellites and Debris

E. Overall Insights

In Fig. 5, the orbital parameters are illustrated. The graphs collectively emphasize the contrast between controlled placement of active satellites and the chaotic nature of space debris. Space debris is concentrated in LEO due to frequent collisions and fragmentation events in this heavily trafficked region. Active satellites exhibit deliberate orbital characteristics optimized for stability and efficiency, whereas debris poses risks due to its unpredictable behavior. These observations underline the importance of effective space traffic management and

debris mitigation strategies to ensure sustainable use of Earth's orbital environment.

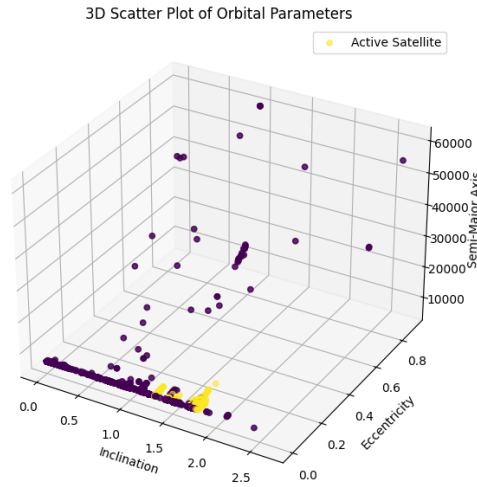


Fig. 5. Orbital Parameters Comparative Variation Map of Satellites and Debris

VII. CONCLUSION AND FUTURE WORK

This study demonstrated the effectiveness of Two-Line Element data in classifying space objects as debris or satellites based on key orbital parameters. The results confirmed that parameters such as eccentricity, inclination, and semi-major axis play a crucial role in classification, while velocity remained largely consistent. A comparative analysis of the FNN, gradient boosting classification and Random Forest models highlighted their advantages and limitations, reinforcing the need for automated classification techniques in space situational awareness. Future research will focus on integrating light curve data, developing an advanced filtering mechanism, and exploring deep learning models to further enhance classification accuracy and real-time space object monitoring. Additionally, collaboration with space agencies and leveraging hybrid machine learning models will be explored to optimize classification performance and support space traffic management efforts.

REFERENCES

- [1] I. Priyadarshini, "Enhanced space debris detection and monitoring using a hybrid bi-lstm-cnn and bayesian optimization," in *Artificial Intelligence and Applications*, vol. 3, no. 1, 2025, pp. 43–55.
- [2] D. Mehrholz, L. Leushacke, W. Flury, R. Jehn, H. Klinkrad, and M. Landgraf, "Detecting, tracking and imaging space debris," *ESA Bulletin(0376-4265)*, no. 109, pp. 128–134, 2002.
- [3] Airbus, "Removedebris - space debris removal," 2025. [Online]. Available: <https://www.airbus.com/en/products-services/space/earth-observation/removedebris>
- [4] H. Klinkrad, *Space debris: models and risk analysis*. Springer Science & Business Media, 2006.
- [5] R. Elkadiri, M. Sultan, A. M. Youssef, T. Elbayoumi, R. Chase, A. B. Bulkhi, and M. M. Al-Katheeri, "A remote sensing-based approach for debris-flow susceptibility assessment using artificial neural networks and logistic regression modeling," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 7, no. 12, pp. 4818–4835, 2014.
- [6] G. Singh, V. Kumar, A. B. Buduru, and S. K. Biswas, "Tracking an untracked space debris after an inelastic collision using physics informed neural network," *Scientific Reports*, vol. 14, no. 1, p. 3350, 2024.
- [7] H. Huang, Y. Wang, Y. Li, Y. Zhou, and Z. Zeng, "Debris-flow susceptibility assessment in china: A comparison between traditional statistical and machine learning methods," *Remote Sensing*, vol. 14, no. 18, p. 4475, 2022.
- [8] A. A. Khan, A. Jamil, D. Hussain, M. Taj, G. Jabeen, and M. K. Malik, "Machine-learning algorithms for mapping debris-covered glaciers: the hunza basin case study," *Ieee Access*, vol. 8, pp. 12 725–12 734, 2020.
- [9] R. Linares, M. K. Jah, J. L. Crassidis, and C. K. Nebelecky, "Space object shape characterization and tracking using light curve and angles data," *Journal of Guidance, Control, and Dynamics*, vol. 37, no. 1, pp. 13–25, 2014.
- [10] T. Kelso, "Celestrak: Satellite tracking and orbital data," 2025. [Online]. Available: <https://www.celertrak.org>
- [11] U. S. S. Force, "Space-track: Satellite catalog and orbital data," 2025. [Online]. Available: <https://www.space-track.org>
- [12] M. KUMAR, K. NAMRATA, and N. Kumar, "Data-driven hyperparameter optimized extreme gradient boosting machine learning model for solar radiation forecasting," *Advances in Electrical & Electronic Engineering*, vol. 20, no. 4, 2022.
- [13] M. Kumar, K. Namrata, and A. Samadhiya, "Deep learning assisted optimal dispatch for renewable-based energy system considering consumer incentive scheme," *Cluster Computing*, vol. 28, no. 4, p. 262, 2025.